

CSE 5830 Homework 3

9/23/2026

Due: 11:59PM, Wednesday 9/30/2026

Show all your work to get a full score.

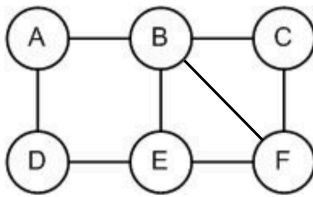
- 1- For a Noisy AND model (Independence of the causal mechanisms when all X s are 1) with binary random variables and leaky parameters $(\lambda_0, \lambda_1, \dots, \lambda_K)$,
 - a) write the expression for $P(Y = 0 | X_1, \dots, X_K)$. (10 points)
 - b) Assume we have a Noisy AND models with 4 parent nodes and leaky parameters of $\lambda_0 = 0, \lambda_1 = 0.9, \lambda_2 = 0.75, \lambda_3 = 0.8, \lambda_4 = 0.85$, what is $P(Y = 0 | X_1 = 1, X_2 = 0, X_3 = 0, X_4 = 1)$? (5 points)
- 2- If $\phi_1(X, Y)$ and $\phi_2(Y, Z)$ are the factors shown below, compute factor $\psi(X, Y, Z) = \phi_1(X, Y) \phi_2(Y, Z)$.
 - a) Draw its table. (10 points)
 - b) What is $\psi(X, Z | Y=1)$? (Draw its table) (5 points)

X	Y	$\phi_1(X, Y)$	Y	Z	$\phi_2(Y, Z)$
1	1	0.8	1	1	0.2
1	2	0.5	1	2	0.2
2	1	0.5	2	1	0.9
2	2	0.6	2	2	1.0

- 3- Consider the pairwise Markov network, $p(\mathbf{x}) = \phi(x_1, x_2) \phi(x_2, x_3) \phi(x_3, x_4) \phi(x_4, x_1)$
 - a) Draw the network that represents the above distribution (2 points).
 - b) Express in terms of ϕ the following: (8 points)

$p(x_1 | x_2, x_4)$
 $p(x_2 | x_1, x_3)$
 $p(x_3 | x_2, x_4)$
 $p(x_4 | x_1, x_3)$

- 4- a) Write all the factors that factorize over the following graph. (6 points)
- b) Consider the graph as a pairwise MN. Write the Gibbs distribution that can be represented by this pairwise MN. (6 points)
- c) Write all the local Markov independencies for the following graph. (6 points)
- d) Are A and F independent given D and C? Why?(2 points)



- 5- Consider the CPD below. (10 points)
- a) What is the probability that $E = e^0$ in the following graph, given an observation $A = a^0, B = b^1, C = c^1, D = d^1$?
- b) What is the probability that $E = e^1$ in the following graph, given an observation $A = a^1, B = b^1, C = c^1, D = d^1$?
- c) What is the probability that $E = e^1$ in the following graph, given an observation $A = a^0, B = b^0, C = c^0, D = d^0$?
- d) Is $E \perp_c B \mid a^0$?
- e) Is $E \perp_c C \mid d^1$?

Note that, for the pairs of probabilities that make up the leaves, the probability on the left is the probability of e^0 , and the probability on the right is the probability of e^1 .

